COMP3242/6242: Deep Learning

Week 7 — Developing, Debugging and Diagnosing Deep Learning Algorithms

Stephen Gould
stephen.gould@anu.edu.au

Australian National University

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Overview

- Numerical Issues
- Algorithmic/Design Issues
- Bugs Can be Hard to Find
- Dataset Issues
- Diagnostics
- Advice for Getting Unstuck
- Tools: Weights & Biases

Simple Arithmetic Quiz

- **Question 1:** What is $255 + 1$?

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import numpy as np

x = np.array([0, 1, 41, 255], dtype='uint8')
x += 1
print(x)
```

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- **Question 2:** What is 16,777,216 + 1?

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- **Question 2:** What is $16{,}777{,}216 + 1$?
- **Answer:** 16,777,216

```
import numpy as np

x = np.array([16777216], dtype='float32')
x += 1
print(x)
```

Pixel Manipulation

- image manipulation (e.g, filtering) is susceptible to these types of underflow and overflow

```
1   import numpy as np
2   import cv2
3
4   img = cv2.cvtColor(cv2.imread(FILE_IN), cv2.COLOR_BGR2RGB)
5   img2 = img + 1
6   cv2.imwrite(FILE_OUT, cv2.cvtColor(img2, cv2.COLOR_RGB2BGR))
7
8   import matplotlib.pyplot as plt
9   plt.figure()
10  plt.subplot(2,1,1)
11  plt.imshow(img); plt.axis('off')
12  plt.subplot(2,1,2)
13  plt.imshow(img2); plt.axis('off')
14  plt.show()
```

Numerical Issues

"anything that can go wrong will go wrong"

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- numerical calculations on a computer are always subject to errors
- deep learning algorithms are full of numerical calculations
- numerical errors can be due to:
  - limited precision arithmetic
    - we just saw some examples
  - algorithmic limitations (e.g., generating true random numbers)
  - careless implementations
    - we will see some example soon
  - bugs!!!

Numerically Stable Standard Deviation

- empirical standard deviation is defined as

$$\hat{\sigma} = \sqrt{\frac{\sum_{i=1}^{n} (x_i - \mu)^2}{n-1}}$$

  where $\mu = \frac{1}{n} \sum_{i=1}^{n} x_i$
- requires two passes through the data

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  where $\mu = \frac{1}{n} \sum_{i=1}^{n} x_i$
- requires two passes through the data
- a seemingly better approach is to perform equivalent calculation

$$\hat{\sigma} = \sqrt{\frac{n \sum_{i=1}^{n} x_i^2 - \left(\sum_{i=1}^{n} x_i\right)^2}{n(n-1)}}$$

  which only requires one pass
- but what can go wrong with this implementation?

Numerically Stable Softmax

- recall the definition of softmax for a vector $z \in \mathbb{R}^K$ is

$$\textbf{softmax}(z) = \left(\frac{\exp z_1}{Z}, \ldots, \frac{\exp z_K}{Z}\right)$$

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- **problem**: susceptible to overflow and underflow

- **solution**: let $z_{\text{max}} = \max\{z_1, \ldots, z_K\}$; then

$$\textbf{softmax}(z) = \left(\frac{\exp(z_1 - z_{\text{max}})}{Z}, \ldots, \frac{\exp(z_K - z_{\text{max}})}{Z}\right)$$

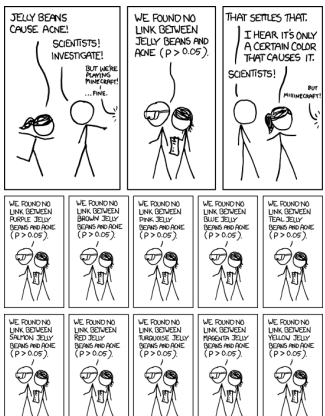
  where $Z = \sum_{k=1}^{K} \exp(z_k - z_{\text{max}})$

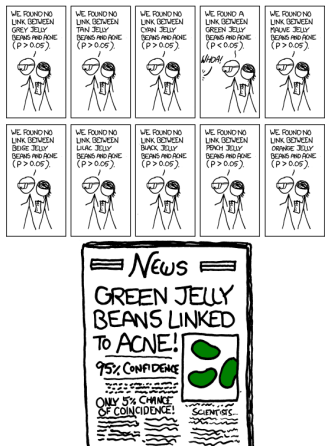
Algorithmic Issues

- poor experiment design (e.g., choosing a favourable random seed)
- systematic error from mini-batch sampling or division into train-test split
- leaking information from test set during training

Misinterpreting Statistical Experiments

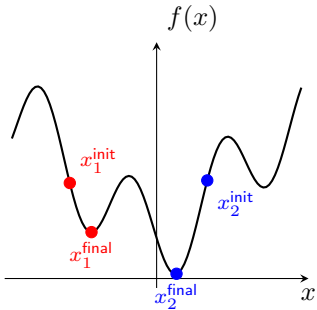
(https://m.xkcd.com/882/)





Favourable Random Seeds

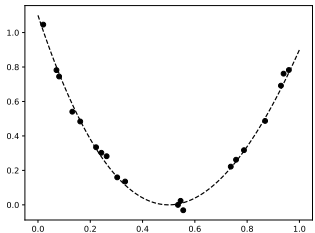
"random chance seems to have operated in our favour"



- gradient descent will take point to its closest local minimum†
- a lucky random seed can favour/penalize one model over another
- always repeat experiments initializing with different random seeds
  - compare different variants of model with the same random seeds

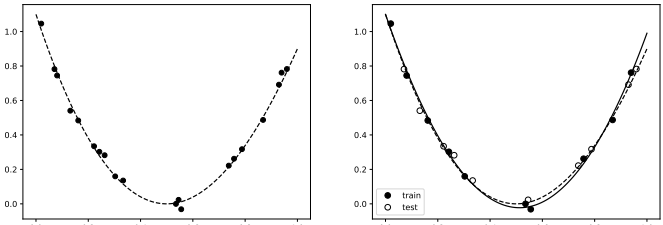
†so long as the function is sufficiently smooth

Sampling Strategies on Regression Example



- training sample distribution should match test sample distribution
  - but avoid correlations between train and test splits, e.g., alternate video frames
- similar effect with stochastic gradient descent if no shuffling between epochs

Sampling Strategies on Regression Example



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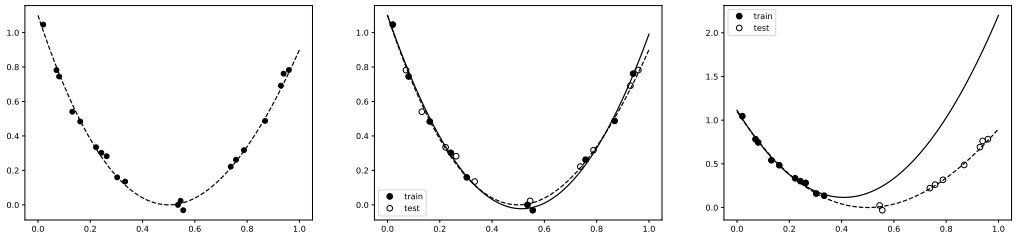

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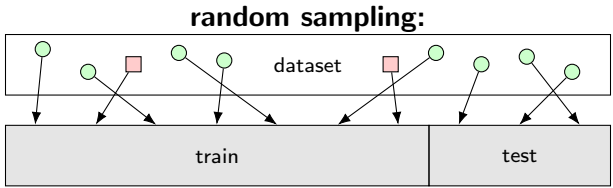
Unbalanced Data

- rare categories may have different distributions between train and test splits



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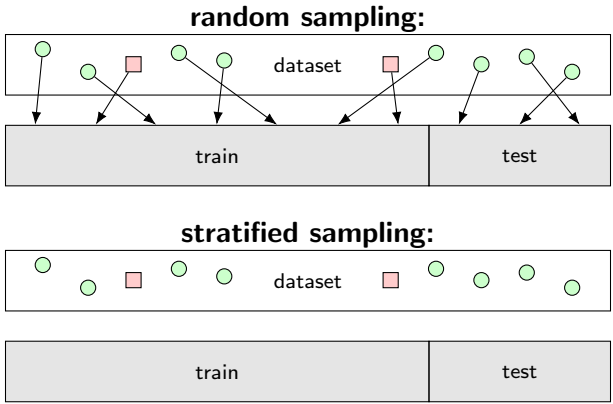

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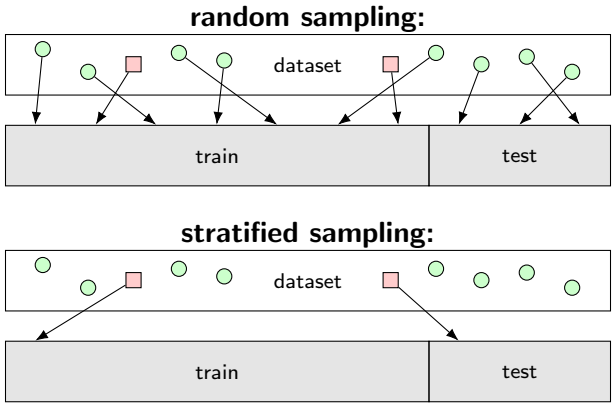

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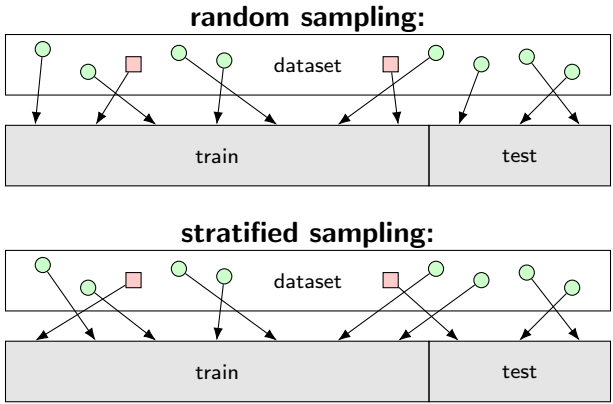

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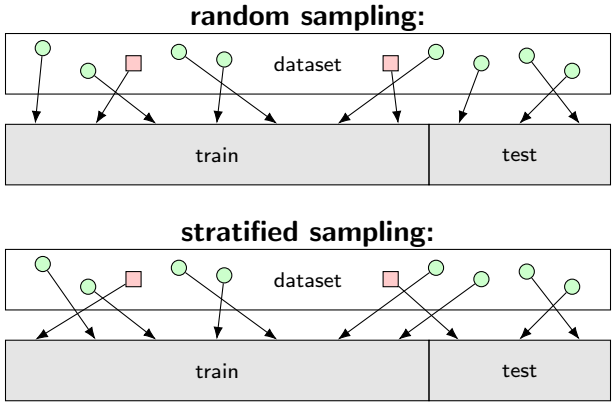

- can also be done in mini-batch sampling
- can also reweight rare samples in loss function

Information Leakage

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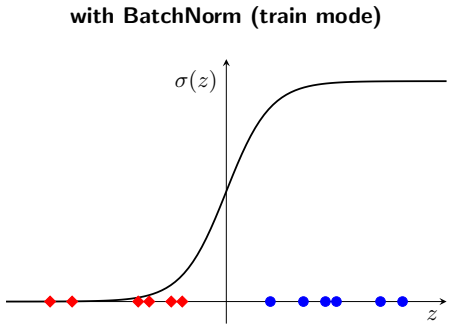
- a common source of information leakage is when pre-processing the data
- another common error is not putting layers into evaluation mode, e.g., `BatchNorm`



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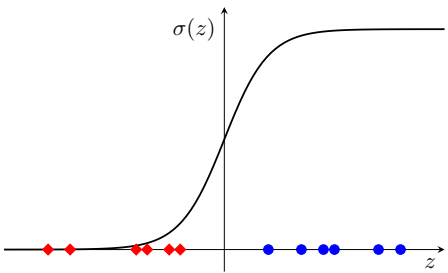
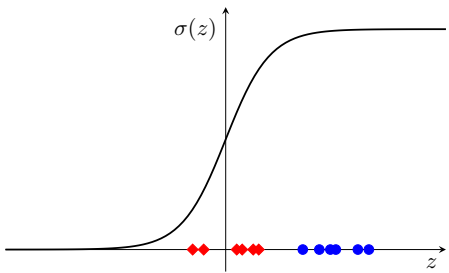
with BatchNorm (train mode)



no BatchNorm (eval mode)



Bugs in Machine Learning Algorithms can be Hard to Find

- consider trying to minimize the following scalar-valued function

$$f(x) = 10x^4 + x^2$$

- we'll use damped Newton's method, which has updates

$$x \leftarrow x - \eta \frac{f'(x)}{f''(x)}$$

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- let's introduce a small bug

correct	**buggy**
$f'(x) = 40x^3 + 2x$	$f'(x) = 40x^3 - 2x$
$f''(x) = 120x^2 + 2$	$f''(x) = 120x^2 - 2$

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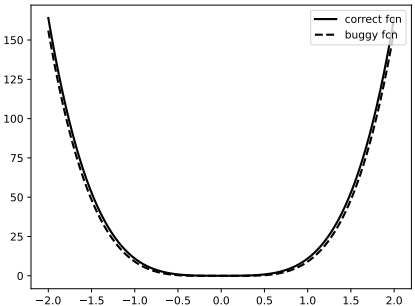

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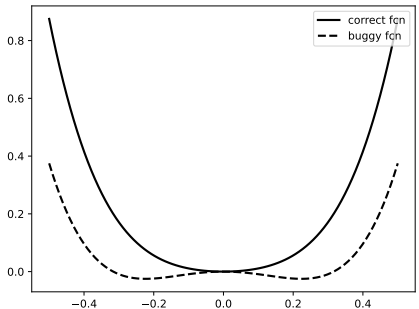

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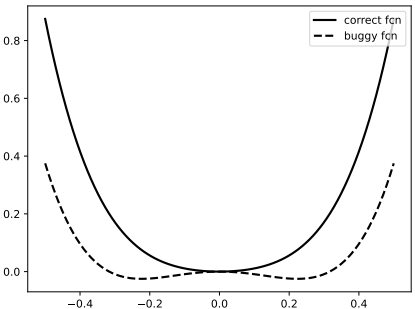

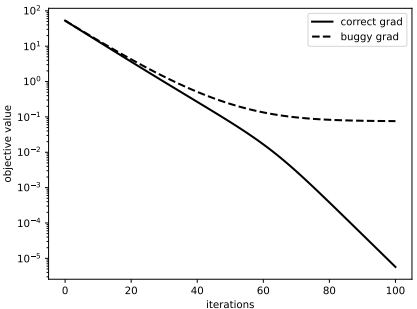

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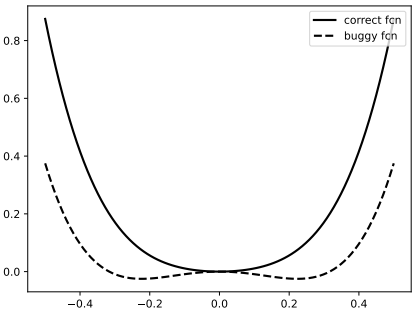
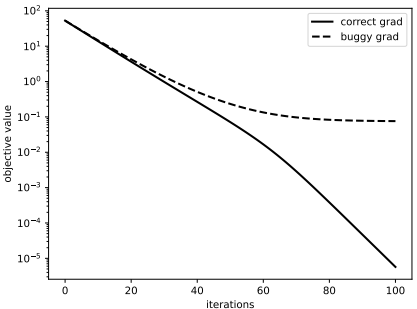


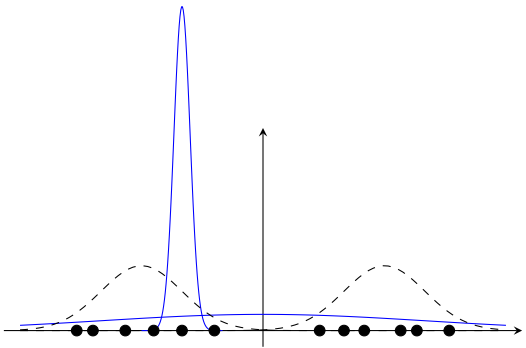


- **take home message:** use automatic differentiation wherever possible

Be Aware of Pathological Cases

- e.g., the true optimal solution of a Guassian mixture model is not desired

$$\text{maximize} \quad \sum_{i=1}^{N} \log \underbrace{\sum_{k=1}^{K} \pi_k \mathcal{N}(x^{(i)}; \mu_k, \Sigma_k)}_{p(x^{(i)};\theta)}$$



Implementation Tips

- **software is written for people, not machines**
- check for `NaN` and `Inf`
- assert pre- and post-conditions
- write simple test cases when debugging (or better yet, before you need to debug)
  - and keep them for future regression testing
- use existing code where possible
  - **deep learning frameworks:** PyTorch, TensorFlow, JAX
  - but don't blindly trust third-party code
  - reproduce published results before extending
- use software revision control (e.g., github)
  - do not store files than can be reproduced or downloaded (but include scripts)
- print out and save debugging information (but not within tight loops)
  - checkpoint models
- use `top`, Task Manager, or `nvidia-smi` to check for memory leaks
- run on small examples before your entire dataset

In case of fire

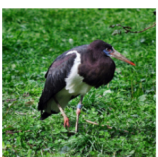
1. git commit
2. git push
3. leave building

Dataset Bias

- **every** dataset is biased!

Noisy/Erroneous Labels

- (almost) **every** dataset is wrong!

MNIST	CIFAR-10	CIFAR-100	Caltech-256	ImageNet	QuickDraw
					
given: 5 corrected: 3	given: cat corrected: frog	given: lobster corrected: crab	given: ewer corrected: teapot	given: white stork corrected: black stork	given: tiger corrected: eye

(source: `https://github.com/cleanlab/cleanlab`)

Diagnostic Tips

"experimental confirmation of a prediction is merely a measurement; experimental disproving of a prediction is a discovery"

- measure everything
- visualize everything
- choice of metrics can influence interpretation of results
  - ask questions of your metrics
- maintain a mental model of your code/algorithm
  - exercise your mental model against your observations
- develop diagnostic tests (next few slides)
- change one thing at a time, i.e., controlled experiments
  - e.g., different components of your loss function
- make sure learnable parameters (weights) are changing
  - make sure weights make a difference—freeze everything else and retrain from initialised weights or perturb during training
- **report trends not single results**
  - this is important not just for diagnoses but also scientific integrity

Loss Functions

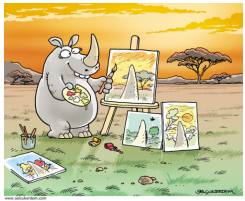
- loss functions are often constructed by summing many terms

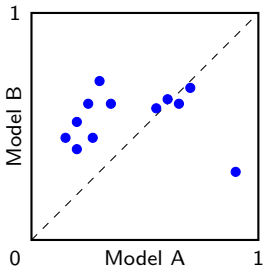
$$L(\theta) = \lambda_1 L_1(\theta) + \lambda_2 L_2(\theta) + \dots$$

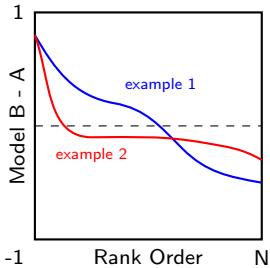
- investigate the contribution of each term by varying the weight of just that term
  - setting the weight to zero should remove the behaviour encouraged by that term
  - setting the weight very high should make that term dominate
- don't just tune hyper-parameters to get best performance, also tune to get better understanding
- not all loss functions have the same scale or are bounded
  - take this into account when setting the weight for each term

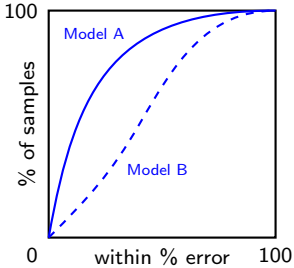
Visualize Your Data

- for high-dimensional features use t-SNE or PCA plots
- to discover correlations or compare models visualize different statistics



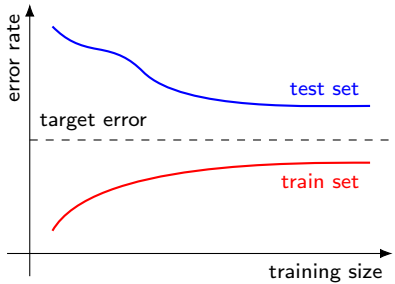




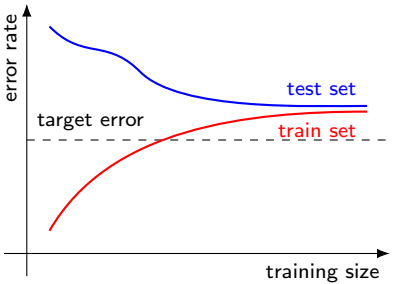


Diagnostic Example: Over-fitting vs. Poor Features

- suppose that our test error is unacceptably high and we suspect the problem is either that the model is **over-fitting** or the **features** are not good enough
  - first hypothesis suggests that training error will be much lower than test error
  - second hypothesis suggests that training error and test error will both be high

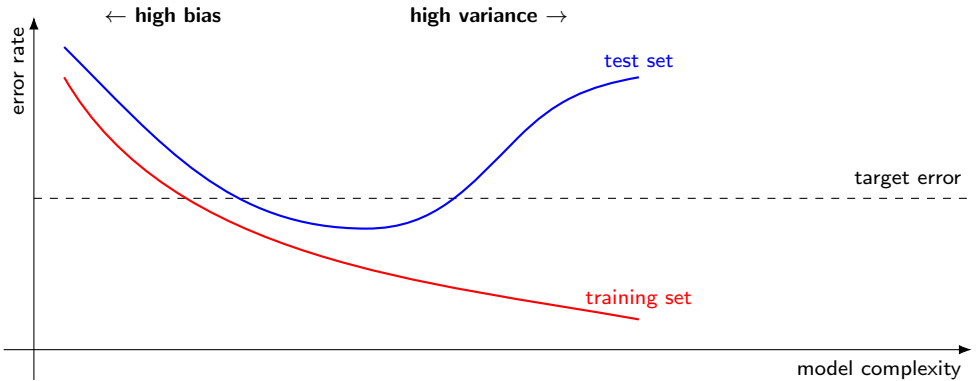


high variance



high bias

Bias/Variance Trade-off



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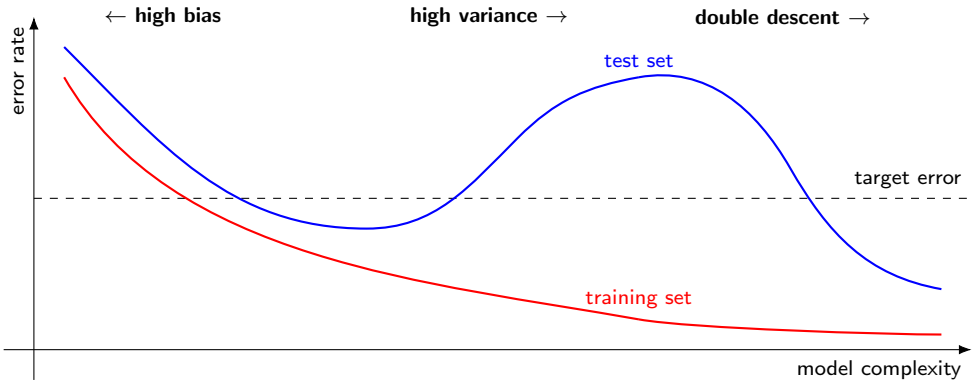
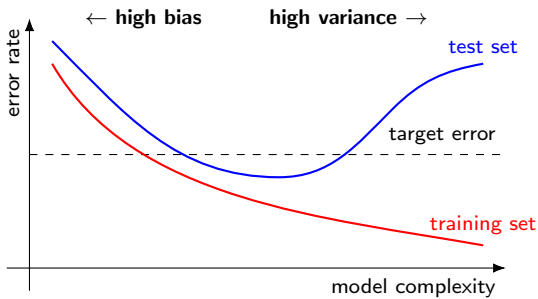


- **double descent** has been observed in deep learning, where further increasing model complexity sometimes results in test set error dropping again
  - not well understood theoretically

Fixes for Bias/Variance Problems

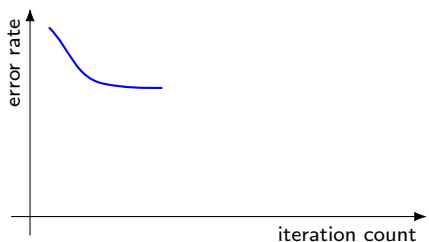
- diagnosing bias and variance problems provides hints as to what to try next
- **for bias problems**:
  - try a larger set of features
  - try a richer model class
- **for variance problems**:
  - try getting more training examples
  - try data augmentation
  - try fewer features
  - try training for longer (and hope for double descent)



Diagnostic Example: Objective vs. Optimization Algorithm

- we may suspect that our poor performance is due to either a problem with our **optimisation algorithm** (e.g., not running for long enough) or a problem with our **objective** (i.e., loss function)

- unfortunately, it is often very difficult to determine whether training has converged



Diagnostic Example: Objective vs. Optimization Algorithm

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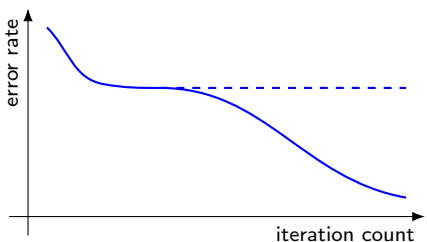


Diagnosing Optimization Issues

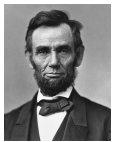
- suppose we care about maximizing some accuracy measure, $J(\theta)$, and our learning algorithm is trying to minimize a surrogate loss, $L(\theta)$
  - let $\theta^\star$ be the parameters returned by the learning algorithm
  - let $\theta^\dagger$ be any other parameters (e.g., guessed or from a different learning algorithm)

	$J(\theta^\dagger) > J(\theta^\star)$	$J(\theta^\dagger) < J(\theta^\star)$
$L(\theta^\star) < L(\theta^\dagger)$	wrong loss	no problem (?)
$L(\theta^\star) > L(\theta^\dagger)$	poor optimization	poor optimization (but got lucky)

Fixes for Optimization/Objective Problems

- diagnosing optimization versus objective problems provides us with hints as to what to try next
- **for optimization problems**:
  - try running for more iterations
  - try using a different algorithm (e.g., AdamW instead of SGD)
  - try random restarts
  - try smoothing (e.g., momentum, exponential moving average)
- **for objective problems**:
  - try different regularization
  - try data augmentation
  - try weighting training examples
  - try a different loss function
  - change the model

Diagnostic Summary



"give me six hours to chop down a tree and I will spend the first four sharpening my axe"

- diagnostics are an important tool when developing your machine learning method
  - we showed examples for bias/variance and optimization/objective but there are many others
  - diagnostics can save a lot of wasted effort by guiding your choice of what to try next
  - they also allow you to develop insights into your particular application and justify your design decisions
- diagnostics often involve repeated experiments with different parameter settings while keeping everything else fixed
  - be organized
- another important diagnostic tool is that of **error analysis**, i.e., understanding where your method is making mistakes, and **ablation analysis**, i.e., understanding which parts of your model help the most

Error Analysis and Ablation Analysis

Error Analysis

- tries to explain the difference between **current** and **perfect** performance
- how much error is due to different model components?
  - plug ground-truth (if available) into each component and see how it affects accuracy
  - add noise to each component and see how it affects accuracy
- does the algorithm fail on a particular subclass of examples?
  - visualize the data and results!

Ablation Analysis

- tries to explain the difference between some **baseline** and **current** performance
- often methods are built in an ad hoc fashion
  - features/components are added in early development and kept even if not needed
- remove features/components from the model one at a time and sees which results in the biggest decrease in performance
  - note that the order of removal matters
- allows simplification of the model, which can then be further developed

Advice for Getting Unstuck

"if you can't solve a problem, then there is an easier problem that you [also] can't solve: find it"

- find a simpler (related) problem
- solve subproblems
- explore boundary conditions
- read papers (both old and new) for ideas
  - be skeptical about results reported in the literature that are not reproducible
- talk to friends and colleagues
  - rubber duck debugging
- introduce more information and slowly remove
  - extreme case: "cheat" by introducing ground-truth features
  - can also test on random features — how should your model perform?
  - generate synthetic test cases

weights & biases

Weights & Biases (wandb)

Weights & Biases (wandb) is the de facto standard tool for:

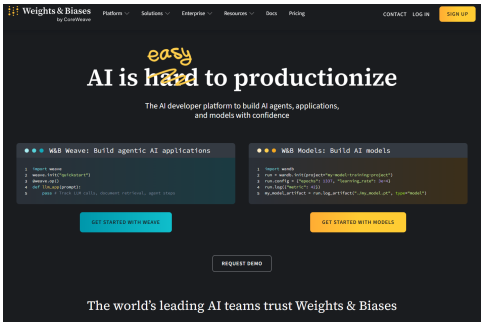
- monitoring training of deep learning models
- streamlining hyperparameter tuning
- logging and recording experiments
- visualizing metrics and internal model features

Requires:

- account on `wandb` server
- local Python package(s)

Not strictly needed for your projects but useful if you're running lots of experiments.

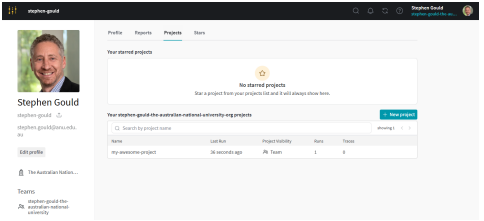
Quick Start

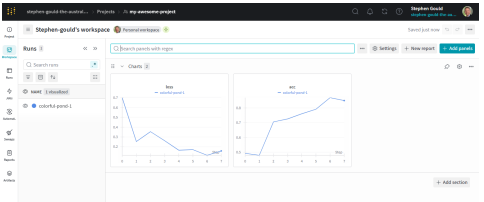


- create (free) account on website
  - `https://wandb.ai`
  - choose academic or personal use
  - choose "W&B Models"
  - choose "Python / PyTorch"
  - generate (and save) API key
- install `wandb` API
  - `pip install wandb`
- cache login credentials
  - `wandb login`
  - paste saved API key

Toy Example: Logging Metrics

```python
from random import random
import wandb

# start a new wandb instance
run = wandb.init(
    entity="...",
    project="my-awesome-project",
    name="first run",
)

# simulate training run
epochs = 10
offset = random() / 5
for epoch in range(2, epochs):
    acc = 1 - 2**-epoch - random() / epoch - offset
    loss = 2**-epoch + random() / epoch + offset

    # log metrics to wandb
    run.log({"acc": acc, "loss": loss})

# finish the run and upload any remaining data
run.finish()
```

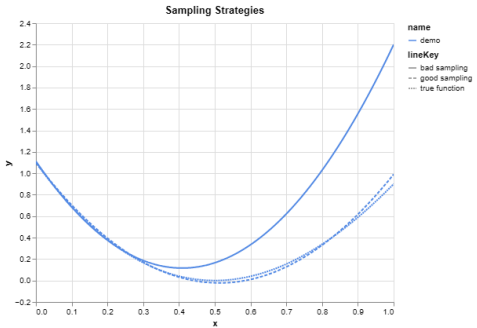




- running multiple times adds additional curves to each plot

Wandb Custom Charts

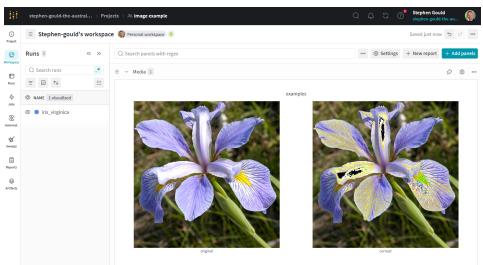
```python
# initialize data
x = np.linspace(0.0, 1.0, 100)
y_true = ...
y_good = ...
y_bad = ...

# log to wandb
wandb.init(project="regression example", name="demo")
chart = wandb.plot.line_series(
    xs=x.T,
    ys=np.vstack((y_true, y_good, y_bad)),
    keys=["true", "good", "bad"],
    title="Sampling Strategies")
wandb.log({"regression_curves": chart})
wandb.finish()
```



- other charts include `confusion_matrix`, `pr_curve`, `scatter`, `bar`

Wandb Images

```python
# load image
img = cv2.imread(BASENAME + ".jpg")
img = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)

# add one to each pixel
img2 = img + 1

# start wandb run and log images
wandb.init(project="image example",
    name="iris_virginica")
with wandb.init() as run:
    run.log({"examples": [
        wandb.Image(img, caption="original"),
        wandb.Image(img2, caption="corrupt"),
        ]})
```



Wandb Checkpointing

- checkpoints allow you to find your best performing model after training
- resume training from some earlier epoch with different setup
  - **danger:** this can make reproducing your results very difficult if you're not careful to document exactly what you've done (e.g., new learning rate, data augmentation, etc)

```
with wandb.init(...) as run:
    # create a model checkpoint
    artifact = wandb.Artifact(name="my_awesome_model", type="model")

    # add model weights to the artifact
    artifact.add_file("model.pt")

    # log the artifact to wandb
    run.log_artifact(artifact, alias=[f"epoch:{epoch}", f"acc:{accuracy}", "best"])
```

```
# resume training from a earlier checkpoint
with wandb.init(...) as run:
    artifact = run.use_artifact("my_awesome_model:best")
    artifact.download()
```

Wandb Hyperparameter Search

- W&B provides a systematic way to try different combinations of hyperparameters and keep track of the results
- involves defining a **sweep** specifying search strategy and metric

```
# configure the sweep
config = {'method': 'grid'}

config['metric'] = {
    'name': 'loss',
    'goal': 'minimize'
}

config['parameters'] = {
    'optimizer' : {'values': ['adam', 'sgd']},
    'fc_layer_size': {'values': [128, 256, 512]},
    'learning_rate': {'distribution': 'uniform', 'min': 1.0e-6, 'max': 1.0e-2},
    'batch_size': {'values': [32, 64, 128]}
}
```

- write your training script to use values from `run.config` (one set of options)
- start the sweep to automate calling training script with different options
  - can be done on multiple machines and repeated multiple times
- see the docs for lots more features and examples